CSCE 46103/56103 Artificial Intelligence – Fall 2026

Machine-Learning Formulation

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This Material's Topics

- Learning from experience and different sources of feedback.
- Inputs, targets, hypotheses, and performance measures.
- Formulating a learning problem before selecting an algorithm.

The Central Question

What should an agent learn, from which experience, and for what purpose?

Learning framework: Russell and Norvig2020, Secs. 19.1–19.2].

From Planning to Learning

Our Earlier MDPs

- A supplied state representation.
- Known transition and reward models.
- A policy computed from those models.

An Agent Learning from Data

- Learn to interpret sensor observations.
- Learn to predict the effects of actions.
- Learn a policy from experience.

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Learning and Planning

An agent can learn transition and reward models from experience and use them to plan. Alternatively, it can learn a policy that maps states directly to actions.

Learning framework: Russell and Norvig2020, Secs. 19.1–19.2].

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Components of a Learning Problem

Component	Question
Task	What should the system do?
Experience	What observations, examples, or interactions are available?
Representation	What information can the model use and produce?
Model	Which candidate mappings or behaviors can be learned?
Performance	How will we judge its behavior on new situations?

Problem formulation: Russell and Norvig2020, Sec. 19.9.1]. Examples adapted for this lecture.

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The algorithm solves the learning problem we specify. Choosing that problem is part of the work.

Problem formulation: Russell and Norvig2020, Sec. 19.9.1]. Examples adapted for this lecture.

Sources of Learning Feedback

Setting	Available feedback	Example
Supervised learning	Input–target examples	Image and object label
Unsupervised learning	Observations without specified target labels	Groups of similar images
Reinforcement learning	Rewards from interaction	Reward for completing a robot task

Learning settings: Russell and Norvig2020, Sec. 19.1]; next-word prediction: Russell and Norvig2020, Sec. 24.2.1].

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Sources of Learning Feedback

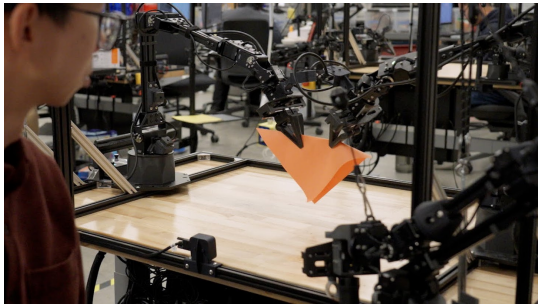
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Self-Supervised Learning

Construct prediction targets from the data itself: use preceding words to predict the next word in a document.

Learning settings: Russell and Norvig2020, Sec. 19.1]; next-word prediction: Russell and Norvig2020, Sec. 24.2.1].

Embodied Agents



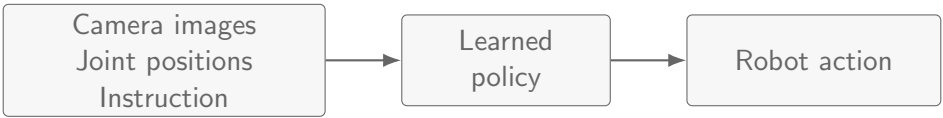
Gemini Robotics Demo: Folding an Origami Fox

- An embodied agent perceives and acts in an environment.
- Physical AI applications include robots interacting with the physical world.
- Gemini Robotics combines visual and language inputs with robot-action outputs.

Example and image: Parada2025].

Robot Learning from Demonstrations

Consider the instruction "Place the object on the plate."



- Demonstrations pair the available inputs with an expert's action.
- The action is the prediction target for imitation learning.
- An observation may omit part of the world; history may also be needed.

Learning Signals in Robotics

Learning task	Experience	Learning signal
Object recognition	Images with named objects	Correct object labels
Action imitation	Expert demonstrations	Expert actions for each input
Reward-based control	Attempts to perform the task	Rewards for outcomes of actions

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Several Learning Problems in One Setting

Embodiment describes the agent's setting. The available learning signal determines how a component can be trained.

Learning settings: Russell and Norvig2020, Sec. 19.1]. Examples are illustrative.

Learning Objectives and Task Success

During Imitation Learning

Predict an action close to the demonstrated action for each recorded input.

During Task Execution

Place the object on the plate despite changes in position, grip, or surroundings.

New initial conditions or disturbances can produce observations absent from the demonstrations.

Teaching discussion of prediction objectives and deployed performance; formulation principles in Russell and Norvig2020, Sec. 19.9.1].

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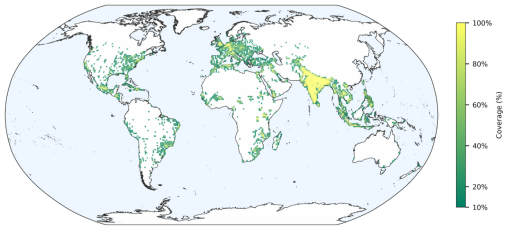
Does perfect action prediction on recorded inputs guarantee task success under new conditions?

No. Measure success on complete task executions as well as action-prediction error on recorded demonstrations.

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Urban Flash-Flood Prediction

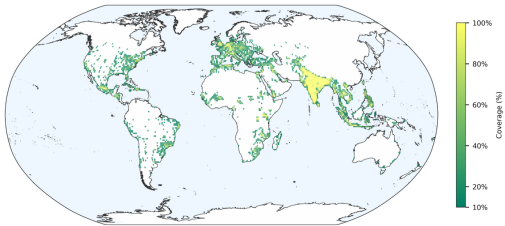


- Inputs: weather data and local geographic attributes.
- Output: risk of a flash flood in the next 24 hours.
- Training labels: historical events extracted from news reports.

System, training-data limitations, and coverage image: Zlydenko and Cohen2026].

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Limits of the Labels

Some floods go unreported. News coverage is denser in cities, affecting which places are represented in the data.

System, training-data limitations, and coverage image: Zlydenko and Cohen2026].

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Inputs and Prediction Targets

For one supervised example, write

$$(x, y) = (\text{available input},\ \text{target output}).$$

A learned hypothesis h produces a prediction $\hat{y} = h(x)$.

Robot Imitation

x: images, joint positions, and instruction.

y: the demonstrated action.

Flood Prediction

x: weather data and local geographic attributes.

y: whether a flash flood was recorded in the following 24 hours.

Learning framework: Russell and Norvig2020, Secs. 19.1–19.2].

Information Available at Prediction Time

A diner arrives at a restaurant. We predict whether the diner will wait for a table.

Candidate feature	**Available then?**
Number of patrons currently present	Yes
Wait estimate displayed by the restaurant	Yes
Actual elapsed time until a table becomes available	No
The diner's eventual decision to wait	No: this is the target

Problem formulation: Russell and Norvig2020, Sec. 19.9.1]. Examples adapted for this lecture.

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Information Leakage

A model evaluated with future information can appear useful while depending on inputs unavailable when a real prediction is needed.

Problem formulation: Russell and Norvig2020, Sec. 19.9.1]. Examples adapted for this lecture.

Classification and Regression

The target determines the supervised-learning task.

Classification

Choose a class from a discrete set.

$$y \in \{\text{yes}, \text{no}\}.$$

Example: will the diner choose to wait?

Regression

Predict a numerical quantity.

$$y \in \mathbb{R}_{\geq 0}.$$

Example: how many minutes until a table is available?

Learning framework: Russell and Norvig2020, Secs. 19.1–19.2].

Representations and Hypothesis Spaces

A hypothesis space $\mathcal{H}$ is the set of candidate models the learner may select.

$$h \in \mathcal{H}, \qquad \hat{y} = h(x).$$

- Features determine what information a model receives.
- The hypothesis space determines which relationships it can express.
- A learning algorithm uses data to choose a hypothesis.

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A Representation Limit

If two diners have identical recorded features but different preferences, a deterministic model using only those features must give them the same prediction.

Learning framework: Russell and Norvig2020, Secs. 19.1–19.2].

Prediction Quality and Practical Value

Prediction Quality

- Classification: compare predicted and observed classes.
- Regression: compare predicted and observed numbers.

Practical Value

- Which mistakes matter most?
- Does the system improve the intended decisions or outcomes?

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A Baseline for Comparison

Compare with a simple reference, such as always predicting the most frequent training class. A useful score needs context.

Problem formulation: Russell and Norvig2020, Sec. 19.9.1]. Examples adapted for this lecture.

Restaurant-Waiting Formulation

We record patrons present, the announced wait estimate, and whether each diner chooses to wait.

1. What are the input x and target y?
2. Is the task classification or regression?
3. Is predicting whether a diner will wait enough to decide whether waiting is better than leaving?

Restaurant example adapted from Russell and Norvig2020, Sec. 19.2]; only two of the original features are retained here.

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One Formulation

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No. The labels record the diner's choices. A recommendation requires comparing the expected benefits and costs of waiting and leaving for that diner.

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A Formulation Checklist

Before choosing an algorithm, specify:

1. The decision or prediction, and when it will be made.
2. The inputs available at that time and the desired output.
3. The source and meaning of training feedback.
4. A suitable representation and hypothesis space.
5. Performance measures, a baseline, and relevant new cases.

Problem formulation: Russell and Norvig2020, Sec. 19.9.1]. Examples adapted for this lecture.

Main Takeaways

- Agents can learn perception, models, or policies from experience.
- The feedback source distinguishes learning settings.
- Inputs, targets, and representations define the prediction problem.
- Accurate prediction and successful decisions require separate checks.

Learning framework: Russell and Norvig2020, Secs. 19.1–19.2].

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Thank You

Any questions?

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